

On-shell Recursion relations at tree-level:

Complex-shifts and Cauchy's thm:

An is characterized by the momenta of the external parts and their type (h_i). $p_i^2 = 0$

$$\sum_i p_i^\mu = 0$$

- Introduce n - $\mathbb{C}$ vectors r_i^μ (some which may be 0)
s.t.

$$(i) \sum_{i=1}^n r_i^\mu = 0$$

$$(ii) r_i \cdot r_j = 0 \quad \forall i, j = 1, \dots, n$$

$$r_i^2 = 0$$

$$(iii) p_i \cdot r_i = 0 \quad \forall i \text{ (no sum)}$$

- We used these to define n -shifted momenta:

$$\hat{p}_i^\mu \equiv p_i^\mu + z r_i^\mu, \quad z \in \mathbb{C}$$

(A) from (i)

$$\sum_{i=1}^n \hat{p}_i^\mu = \sum_{i=1}^n p_i^\mu + z \sum_{i=1}^n r_i^\mu = 0 \quad \left| \sum_{i=1}^n \hat{p}_i^\mu = 0 \right.$$

(B) By (ii) and (iii)

$$\hat{p}_i^2 = (p_i^\mu + z r_i^\mu)(p_i^\mu + z r_i^\mu)$$

$$= p_i^2 + z^2 r_i^2 + 2z r_i^\mu p_i^\mu = 0$$

$\hat{p}_i$ is on shell

$$\left| \hat{p}_i^2 = 0 \right.$$

(C) (Non-trivial: at least two and no more than $(n-2)$ p_i)
s.t. $p_i^2 \neq 0$

For a non-trivial subset of generic momenta

$\{p_i\}_{i \in I}$, define

$$P_I^\mu = \sum_{i \in I} p_i^\mu$$

then $\hat{P}^2$ is linear in z :

$$\hat{P}_I^2 = \left(\sum_{i \in I} \hat{p}_i \right)^2 = P_I^2 + 2Z P_I \cdot R_I$$

$$R_I = \sum_{i \in I} r_i$$

Define: $\boxed{z_I \equiv -P_I^2 / 2P_I \cdot R_I}$

$$\Rightarrow 2P_I R_I = -P_I^2 / z_I$$

$$\hat{P}_I^2 = P_I^2 - (P_I^2 / z_I) z$$

$$= (P_I^2 / z_I) (z_I - z) \quad \text{or}$$

$$\boxed{\hat{P}_I^2 = -\left(\frac{P_I^2}{z_I}\right)(z - z_I)}$$

* From (A) and (B) we can consider our amp A_n in terms of shifted momenta $\hat{p}_i^H$.

$\hat{A}_n(z)$: shifted amplitude | Holomorphic f^n

$$\boxed{A_n = \hat{A}_n(z)|_{z=0}}$$

- A_n has no branch cuts, A_n is a rational f^n of $\langle \rangle$ and $[]$.

and Analytic structure captured by its poles: physical particles.

$\hat{A}_n(z)$ has simple poles $\sim \frac{1}{z}$ in the z -plane

- A_n can have poles from $1/\hat{P}_I^2$ but

$$\frac{1}{\hat{P}_I^2} = -\frac{z_I}{P_I^2(z - z_I)}$$

So we have a simple pole at z_I . Further,

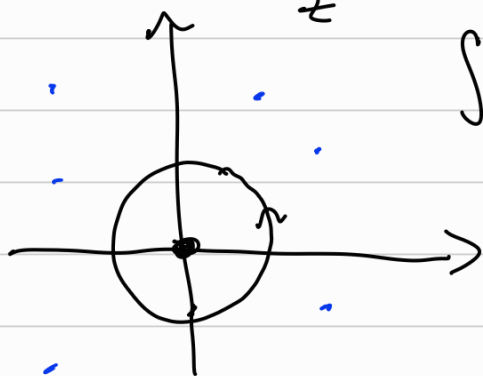
since $P_I \neq 0$ and $z_I \propto P_I$ then

$$z_I \neq 0$$

- Thus, $\hat{A}_n(z)$ has simple poles away from the origin ($z_I \neq 0$).

→ We are assuming LOCALITY: the amplitudes can be derived from some local Lagrangian so that propagators determine the poles

- Study the loop $\oint_C \frac{\hat{A}(z)}{z}$ in the complex z plane



$$\begin{aligned} \int dz \left(\frac{\hat{A}(z)}{z} \right) &= 2\pi i \sum_{res} \hat{A}(z) \Big|_{z=0} \\ &= 2\pi i \hat{A}_n(0) \\ &= 2\pi i A_n \end{aligned}$$

If we deform the contour to include all



$$\frac{1}{2\pi i} \oint_C dz \frac{\hat{A}_n(z)}{z} = - \sum_{z_I} \text{Res}_{z=z_I} \left(\frac{\hat{A}_n(z)}{z} \right) - A_n + B_n$$

Since all extensions have opposite contour orientation...

Not sure why this is exactly the answer...

$$A_n = - \sum_{z_I} \text{Res}_{z=z_I} \left(\frac{\hat{A}_n(z)}{z} \right) + B_n$$

B_n : Residue at $z = \infty$ $z \rightarrow \frac{1}{w}$

$$\frac{\hat{A}_n(z)}{z} = \sum_{n=-\infty}^{\infty} a_n z^n / z = \sum_{n=-\infty}^{\infty} a_n z^{n-1}$$

$$z \rightarrow \frac{1}{w} \Rightarrow \sum_{n=-\infty}^{\infty} a_n w^{1-n} \Rightarrow$$

- At a z_I -pole prop. $1/\hat{p}_I^2$ goes on-shell. In the limit $z \rightarrow z_I$ factorizes into two on-shell parts:

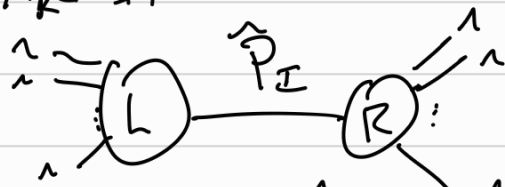
$$\hat{A}_n(z) \xrightarrow{z \rightarrow z_I} \hat{A}_L(z_I) \left(\frac{1}{\hat{p}_I^2} \right) \hat{A}_R(z_I) = \frac{z_I}{(z - z_I)} \hat{A}_L(z_I) \left(\frac{1}{\hat{p}_I^2} \right) \hat{A}_R(z_I)$$

Thus, $-\text{Res}_{z=z_I} \left(\frac{\hat{A}_n(z)}{z} \right) = -\text{Res}_{z=z_I} \left(\frac{z_I}{z - z_I} \right) \hat{A}_L(z_I) \frac{1}{\hat{p}_I^2} \hat{A}_R(z_I)$

$$= \text{Res}_{z=z_I} \left(\frac{z_I}{z(z-z_I)} \right) \hat{A}_L(z_I) \left(\frac{1}{P_I^2} \right) \hat{A}_R(z_I)$$

$$z=0 : \rightarrow A_L (P_I^2)^{-1} A_R$$

$$z=z_I : A_L (P_I^2)^{-1} A_R$$



$$\text{Res} \left(\frac{\hat{A}_I(z)}{z} \right) \Big|_{z=0} = - \sum_a \text{Res} \left(\frac{\hat{A}_I(z)}{z} \right) \Big|_{z=z_a} = \text{Res} \left(\frac{\hat{A}_I(z)}{z} \right) \Big|_{z=z_a}$$

$$A_n = - \sum_a B_n$$

* I'm not sure if (-B_n) is then a Typo...

* Recall from QFT that at 1-loop we have $\int \frac{d^4 k}{(2\pi)^4} \left\{ \frac{-i}{k^2 + m^2 + \Sigma - i\epsilon} \right\} e^{ik(x_1 - x_2)}$

∴ using $\frac{1}{k+a}$ expanded at $a=0$

$$\frac{1}{k+a} = \frac{1}{k} - \frac{a}{k^2} + \frac{a^2}{k^3} + \dots$$

$$\Rightarrow \frac{1}{P_I^2 + 2z P_I R_I} \approx \frac{1}{P_I^2} - \frac{2z P_I R_I}{(P_I^2)^2} + \dots$$

$$\approx \frac{1}{P_I^2} + \frac{z_I}{(P_I)^2} = \frac{1}{(P_I)^2} \left(1 + \frac{z}{z_I} \right) = \frac{z_I + z}{P_I^2 z_I}$$

$$\frac{1}{P_I^2} \left(\frac{1}{1 + z \left(\frac{2P_I R_I}{P_I^2} \right)} \right) = \frac{1}{P_I^2} \left(\frac{1}{1 - \frac{z}{z_I}} \right)$$

$$= P_I \dots \dots \text{NOPE} \dots \dots$$

* The momentum of the internal line is on-shell $\hat{P}_I^2 = 0$, while (L) and (R) represent on-shell amplitudes evaluated at $z = z_I$ (These are Subamplitudes)

Rules: Write the scalar prop $\frac{1}{P_I^2}$ of the unshifted momenta.

* We assume: $B_n = 0$

$$\hat{A}_n(z) \rightarrow \text{for } z \rightarrow \infty$$

i.e. the amplitude has compact support.

→ If $\hat{A}_n(z)$ has compact supp, then we say our extension $\hat{p}_i^\mu = p_i^\mu + z r_i^\mu \Rightarrow$ VALID

• For a valid shift, the n -pt on-shell amp. is completely determined in terms of lower pt on-shell amps:

$$\Delta_n = \sum_{\text{diagrams}} \hat{A}_L(z_F) \frac{1}{P_F^2} \hat{A}_R(z_F) = \sum_{\text{diagrams}} \hat{A}_L \xrightarrow{\hat{P}_F} \hat{A}_R$$

BCFW Recursion Relations:

• $r_i^\mu = 0$ might occur.

BCFW: exactly two lines, say i and j are selected as the only ones with non-vanishing shift vectors.

* In $D=4$:

$$\begin{aligned} |\hat{i}\rangle &\equiv |i\rangle + z|j\rangle, & |\hat{j}\rangle &= |j\rangle \\ |\hat{i}\rangle &= |i\rangle, & |\hat{j}\rangle &= |j\rangle - z|i\rangle \end{aligned}$$

$$\hat{p}_i^\mu = p_i^\mu + z r_i^\mu \quad | [i, j] \rangle - \text{shift}$$

$$\hat{p}_i^{\alpha\dot{\alpha}} = -|i\rangle\langle i|$$

$$\begin{aligned} (\hat{p}_i)_{\alpha\dot{\alpha}} &= -|i\rangle\langle i| = z|j\rangle\langle i| \\ &= (-|i\rangle - z|j\rangle)\langle i| \\ &= -(|i\rangle + z|j\rangle)\langle i| \\ &= -|\hat{i}\rangle\langle \hat{i}| \end{aligned}$$

$$\Rightarrow \text{Thus } |\hat{i}\rangle = |i\rangle + z|j\rangle, \quad \langle i| \\ r_i^\mu = \Rightarrow (r_i)_{\alpha\dot{\alpha}} = -|j\rangle\langle i|$$

$$\hat{r}_{ij}^{\alpha\alpha} = -|j\rangle\langle i|_{\alpha} = -\lambda_{ja}\lambda_{ia}$$

$$\hat{P}_j^{\alpha\alpha} = -|j\rangle\langle j| - z|j\rangle\langle j|$$

$$= -(|j\rangle + z|j\rangle)\langle j|$$

$$\hat{P}_j^{\alpha\alpha} = -(|j\rangle + z|j\rangle)\langle j|$$

$$= -|j\rangle\langle j|$$

$$\hat{r}_j^{\alpha\alpha} = -|j\rangle\langle j|$$

Note $\hat{r}_j\hat{r}_j = 0$

$$\hat{r}_j^{\beta\beta} \in_{\beta\alpha} \in_{\beta\alpha} = +|j\rangle\langle j|_{\alpha}$$

$$\hat{r}_j + \hat{r}_i \neq 0$$

* For the above shift we find:

$$A_n = \sum_{\alpha, i} \hat{A}_L(z_i) \left(\frac{1}{P_L^2} \right) \hat{A}_R(z_i) = \sum_{\text{diagrams}} \hat{i} \rightarrow \text{L} \xrightarrow{\hat{P}_L} \text{R} \leftarrow \hat{j}$$

Sum over all channels I s.t. the shifted lines i, j are on opp. sides of the fac. diagram

• We note: $[\hat{j}\hat{i}] = \langle ij \rangle$

$$[\hat{i}\hat{j}] = \langle ij \rangle$$

• So, from above we concluded that

$$[\hat{i}] = \langle i|$$

$$[\hat{j}] = |j\rangle$$

Furthermore, we can not say much about $[\hat{i}]$ aside from that $[\hat{i}\hat{k}] \neq i, i$

is linear in z . (same for $\langle 1|$)

* Consider the Parke-Taylor Amplitude:

$$\Delta_n[1^- 2^- 3^+ \dots n^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

(i) Take shifts on 1^- and 2^+

$$\langle 1^- | = \langle 1 |, \quad |\hat{2}^+ \rangle = |2^+ \rangle - z |1^+ \rangle$$

$$\frac{\langle 1\hat{2} \rangle \langle \hat{2}3 \rangle + z \langle 13 \rangle}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle}$$

For large z the second term dominates so Δ_n then falls off as $1/z \dots$

$\Rightarrow$ Let $\underline{1^-} = i, \quad \underline{5^+} = j$

$$\frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 56 \rangle \dots}$$

$$\rightarrow \langle 12 \rangle^2 / (\langle 12 \rangle \langle 23 \rangle \dots (\langle 45 \rangle + \langle 41 \rangle z) (\langle 56 \rangle + z))$$

Here we get terms up to z^2

$\Rightarrow$ Finally, we note if our choice are adjacent

$$i=4, j=5$$

$$\langle 12 \rangle^4 / \langle 12 \rangle \dots \langle 34 \rangle \langle 45 \rangle \langle 56 \rangle$$

$$|\hat{4} \rangle = |4 \rangle$$

$$|\hat{5} \rangle = |5 \rangle - z |4 \rangle$$

$$\langle 34 \rangle \langle 45 \rangle \langle 56 \rangle = \langle 34 \rangle \langle 45 \rangle (\langle \hat{5}6 \rangle) \sim (1/z) (a + z)$$

which goes like $1/z$.

Non-adjacent $3 \quad 5$

$$\langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \langle 56 \rangle$$

$$\langle 23 \rangle \langle 34 \rangle \langle 4 | (|5 \rangle - z |3 \rangle) (z \langle 51 \rangle + \langle 31 \rangle) \langle 6 \rangle \dots$$

so again... this goes like $\frac{1}{z^2 + z + 1}$

ex 3.3) $[1, 2^-]$ - shift

$$\begin{aligned} |1\rangle &= |1\rangle + z |2\rangle & |1\rangle &= |1\rangle \\ |2\rangle &= |2\rangle - z |1\rangle & |2\rangle &= |2\rangle \end{aligned}$$

$$\langle 12 \rangle^4$$

$$A_n[123 \dots n] = \langle 12 \rangle \langle 23 \rangle + \dots \langle n1 \rangle$$

$$= - \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} = - \frac{\langle 12 \rangle^4}{\langle 12 \rangle (\langle 23 \rangle + z \langle 31 \rangle) \dots \langle n1 \rangle}$$

Pole at $z = \langle 31 \rangle$

$$\hat{A}_n(z) = \frac{\langle 12 \rangle^4}{\langle 12 \rangle (\langle 23 \rangle + z \langle 31 \rangle) \dots \langle n1 \rangle} =: \alpha / (1 - z \frac{\langle 13 \rangle}{\langle 23 \rangle})$$

$$\hat{A}_n(z) = \alpha \left(1 + z \frac{\langle 13 \rangle}{\langle 23 \rangle} + \left(z \frac{\langle 13 \rangle}{\langle 23 \rangle} \right)^2 + \dots \right)$$

$$\text{Res}\left(\frac{A_n}{z}\right) = \alpha \left(\frac{1}{z} + \frac{\langle 13 \rangle}{\langle 23 \rangle} + \dots z + \dots \right)$$

Thus, the residue $a_{-1} = \alpha$ is given by

$$\frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \dots \langle n1 \rangle} = A_n$$

so the Eq.

$$A_n = - \text{Res}\left(\frac{\hat{A}_n(z)}{z}\right)$$

means im off by a - sign

and Try for a $[1, 3]$ -shift

$$\langle 12 \rangle^4$$

$$\hat{A}(z) = \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \dots \langle n1 \rangle$$

$$\sim (\langle 23 \rangle - \langle 21 \rangle z) (-1) (\langle 43 \rangle - \langle 41 \rangle z) \dots \langle n1 \rangle$$

$$\sim (\langle 23 \rangle - \langle 21 \rangle z) (\langle 34 \rangle - \langle 14 \rangle z) \dots$$

So we establish two poles

I get something of the form

$$\text{Res}\left(\frac{\hat{A}}{z}\right) \sim \frac{A_n}{\langle 24 \rangle \langle 13 \rangle} (\langle 14 \rangle^2 - \langle 12 \rangle^2) \dots$$

Not sure what to do about that

ex 3.4 $E_{-}^{\mu}(p) = - \frac{\langle p | \gamma^{\mu} | q \rangle}{[q p]}$

If we shift p
 $\hat{E}_{-}^{\mu}(\hat{p}) = - \frac{\langle \hat{p} | \gamma^{\mu} | q \rangle}{[q \hat{p}]}$

$p = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$

$|i\rangle = |i\rangle + z |j\rangle$
 $|\hat{i}\rangle = |i\rangle$

$E_{-}^{\mu}(\hat{p}) = - \frac{\langle \hat{p} | \gamma^{\mu} | q \rangle}{\sqrt{2}([q p] + \bar{z}[q j])}$

$E_{+}^{\mu}(\hat{p}) = - \frac{\langle q | \gamma^{\mu} | \hat{p} \rangle}{\sqrt{2} \langle q \hat{p} \rangle} = \frac{\langle q | \gamma^{\mu} | p \rangle + \bar{z} \langle q | \gamma^{\mu} | j \rangle}{\langle q p \rangle}$

$E_{-}^{\mu}(\hat{j}) = - \frac{\langle \hat{j} | \gamma^{\mu} | q_j \rangle}{\sqrt{2} [q_j \hat{j}]} = - \frac{(\langle j | \gamma^{\mu} | q_j \rangle - \langle i | \gamma^{\mu} | q_j \rangle)}{[q_j j]}$

I think the point is we can always shift q "oppositely":

$[q p] \left(1 + z \frac{[q j]}{[q p]} \right)$
 $[q] = [p] - z [j]$ *

$1 + z \left(\frac{[p j]}{-z [j p]} \right) = 2$
 $\frac{\langle p | \gamma^{\mu} | p \rangle - z \langle p | \gamma^{\mu} | j \rangle}{-z [j p]} = 2$

$-\frac{\langle p | \gamma^{\mu} | j \rangle}{\sqrt{2} z [j p]}$

~ effectively makes $q \rightarrow j$
 For E_{+} : $q = j$ solves that as well.
 * we can get rid of the explicit extra term by making $q = j$ and vice versa.
 In general, it makes sense this won't play a role.

* For Color-ordered gluon tree Amp. under a BCFW shift of adjacent lines i and j

$$\boxed{\begin{array}{ccccc} |i, j\rangle & [-, -\rangle & [-, +\rangle & [+, +\rangle & [+, -\rangle \\ \hat{A}_n(z) \sim & 1/z & 1/z & 1/z & z^3 \end{array}}$$

* If i and j are NOT adjacent then we gain an extra power of $1/z$ in each case as found above!

* I did not try $[+, -\rangle$ so let me do that now $[3, 1)\rangle$:

$$\begin{aligned} |3\rangle &= |3\rangle + z|2\rangle, \quad |1\rangle = |1\rangle - z|3\rangle \\ \hat{A}_n(z) &= \frac{(\langle 21\rangle - z\langle 23\rangle)^3}{\langle 23\rangle \dots (\langle n1\rangle - \langle n3\rangle z)} \end{aligned}$$

$$\sim z^3 \alpha + z^2 \beta$$

Proof of Parke-Taylor formula:

- For n , suppose PT is true for all $(n-1)$ cases then write the recursion relation for $A_n[1^+ \dots n^+]$

Based on a $[1, 2)\rangle$ -shift

$$\begin{aligned} A_n[1^+ 2^- 3^+ \dots n^+] &= \sum_{k=4}^n \sum_{n^+}^{\hat{z}^-} \hat{A}_{n-(k-3)}[\hat{1}^-, \hat{P}_I^{h_I}, k^+, \dots, n^+] \frac{1}{P_I^2} \\ &\quad A_{k-1}[-P_I^{h_I}, \hat{2}^-, 3^+, \dots, (k+1)^+] \end{aligned}$$

- Internal momenta P_I , $k=4, \dots, n$

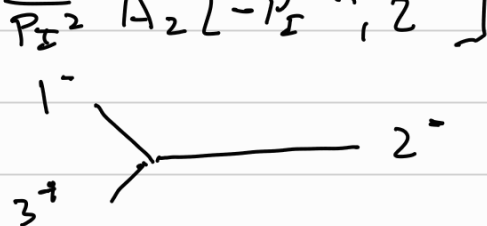
$$P_I = p_2 + p_3 + \dots + p_{k-1}$$

Before $\pm$ were on: $n^+ = 3^+$, $k=3$

Internal momenta:

$$A_{(3-3+3)}[\hat{1}^-, \hat{P}_I^{h_I}, 3^+]$$

$$P_I = P_2 + P_3 \dots$$



$$\text{while } \hat{P}_I = \hat{P}_2 + P_3 + \dots + P_{n-1}$$

* Since $A[-+ \dots +]$ variables except for $n=3$

so, let us sum over helicity first

$$\Lambda_n[1^- 2^- 3^+ \dots n^+] = \sum_{k=4}^n \left(\hat{A}_{n-k+1}[\hat{1}^-, \hat{P}_2^+, k^+ \dots n^+] \frac{1}{P_I^2} \hat{A}_{k-1}[-\hat{P}_I^-, \hat{2}^-, 3^+ \dots (k-1)^+] + \hat{A}_{n-k+1}[\hat{1}^-, \hat{P}_I^-, k^+, \dots, n^+] \frac{1}{P_I^2} \hat{A}_{k-1}[-\hat{P}_I^+, \hat{2}^+, 3^+ \dots] \right)$$

* For the first sum

$\rightarrow \hat{A}_{n-k+1}[\hat{1}^-, \hat{P}_2^+, k^+ \dots n^+]$ can be at most Λ_3

meaning the only term contributing to this is

given by $k=n$

* For the second $k-1=3$ so $k=4$

$$\Lambda_n[1^- 2^- 3^+ \dots n^+] = + \hat{A}_3[1^-, \hat{P}_I^+, n^+] \frac{1}{P_I^2} \hat{A}_{n-1}[-\hat{P}_I^-, 2^-, 3^+, \dots, (n-1)^+] + \hat{A}_{n-1}[\hat{1}^-, \hat{P}_I^-, 4^+, \dots, n^+] \frac{1}{P_I^2} \hat{A}_3[-\hat{P}_I^+, \hat{2}^+, 3^+]$$

Diagrammatically: $\Lambda_n =$

* $\hat{P}_I$ is evaluated at the residue $z = z_{\pm}$ s.t. $\hat{P}_I^2 = 0$

First diagram: $P_I = P_2 + \dots + P_{n-1}$

and so $P_I = -(P_1 + P_n)$

$$\Rightarrow \hat{P}_I = -(\hat{P}_1 + P_n)$$

* Define $P_{ij} \equiv P_i + P_j$

then $\hat{P}_I^+ = -\hat{P}_{1n}^+$

$$(\text{first term}) = \hat{A}_3[1^-, -\hat{P}_{1n}^+, n^+] \frac{1}{P_I^2} \hat{A}_{n-1}[\hat{P}_{1n}^-, \hat{2}^-, 3^+, \dots, (n-1)^+]$$

Second term:

$$P_I = P_2 + P_3 \Rightarrow \hat{P}_I^+ = \hat{P}_2 + P_3$$

$$\hat{P}_I^+ = \hat{P}_{23}^+$$

$\langle 123 \rangle^3$

$$\Delta_n [i23^+] = \langle 12 \rangle \langle 23 \rangle$$

flip helicity $\langle \rangle \rightarrow []$ and

By cyclicity:

$$\hat{\Delta}_3 [\hat{1}^+, -\hat{p}_{in}^+, n^+, \hat{1}^-] = \hat{\Delta}_3 [-\hat{p}_{in}^+, n^+, \hat{1}^-]$$

$$= \frac{[-p_{in}^+ n]^3}{[n \hat{1}] [\hat{1} \hat{p}_{in}]}$$

where we use

$$|-p\rangle = -|p\rangle \quad \text{and} \quad |-p] = +|p]$$

$$\hat{p}_{in} = \hat{p}_1 + p_n \Rightarrow \hat{p}_{12}^2 = 0 \stackrel{\text{on-shell}}{=} 2 \hat{p}_1 \cdot p_n = \langle 1n \rangle [\hat{1} n]$$

* For generic momenta the only way for this to vanish is to take z s.t.

$$[\hat{1} n] = 0 \quad (\text{choice of } z)$$

This would mean $\Delta_3 [\hat{1}^+, -\hat{p}_{in}^+, n^+]$ has a vanishing denominator. Note, however

$$\langle \hat{p}_{in} | [\hat{p}_{in}] | n \rangle = -\langle \hat{p}_{in} | n \rangle = -(\langle \hat{p}_1 + p_n | n \rangle) = -\langle 1n \rangle$$

$$= 0$$

and since $|\hat{p}_{in}\rangle \neq 0$

$$[\hat{p}_{in} n] = 0$$

$\Rightarrow$ Similarly, $[\hat{1} \hat{p}_{in}] = 0 \Rightarrow \hat{\Delta}_3 \rightarrow 0$ on-shell

$$\hat{\Delta}_3 [\hat{1}^-, \hat{p}_{in}^+, n^+] = 0$$

Only left with the second term

$$\Delta_n [1^- 2^- 3^+ \dots n^+] = \frac{\hat{1}^-}{u^+} \textcircled{1} \xrightarrow{\hat{p}_2} \textcircled{2} \xrightarrow{\hat{2}^+}{3^+}$$

$$= \hat{\Delta}_{n-1} [\hat{1}^-, \hat{p}_{23}^-, u^+, \dots, n^+] \frac{1}{p_{23}^2} \hat{\Delta}_3 [-\hat{p}_{23}, \hat{2}^+, 3^+]$$

Inductive step: Assume $\hat{\Delta}_{n-1}$ holds

$$\Delta_n = \left(\frac{\langle \hat{1} \hat{p}_{23} \rangle^4}{\langle \hat{1} \hat{p}_{23} \rangle \langle \hat{p}_{23} u \rangle \langle u n \rangle \dots \langle n \hat{1} \rangle} \right) \left(\frac{1}{\langle 23 \rangle \langle 22 \rangle} \right) \left(\frac{[3 \hat{p}_{23}]^3}{[\hat{p}_{23} \hat{2}] [\hat{2} 3]} \right)$$

Now

$$\langle 1 \hat{p}_{23} \rangle [3 \hat{p}_{23}] = -\langle \hat{1} \hat{p} \rangle [\hat{p} 3] = \langle \hat{1} | \hat{p}_{23} | 3 \rangle = \langle \hat{1} | (\hat{2} + 3) | 3 \rangle$$

$$B_{\text{of}} \langle \hat{1} \hat{2} \rangle = \langle 12 \rangle, \quad [\hat{2} 1] = [21] \quad \text{so}$$

numerator:

$$\Delta_n = \frac{-\langle 12 \rangle^3 [23]^3}{()}$$

$$\begin{aligned} \text{Next: } \langle \hat{P}_{23} 4 \rangle [\hat{P}_{23} \hat{2}] &= \langle 4 | (\hat{2} + 3) | \hat{2} \rangle = \langle 4 | 3 | \hat{2} \rangle \\ &= -\langle 43 \rangle [32] \\ &= -\langle 34 \rangle [23] \end{aligned}$$

$$\therefore \boxed{\Delta_n = \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 45 \rangle \dots \langle n1 \rangle} \quad \langle \hat{2} \rangle = 12 \rangle$$

□
Q.E.D.

NMHU amplitudes: Consider the 5pt example

$$A_5[1^- 2^- 3^+ 4^+ 5^+] = A_5[1^+ 2^+ 3^- 4^- 5^-] = \frac{[145]^4}{[12][23][34][45][51]}$$

$[i, j]$ shifts are for helicities where
 $|\hat{i}\rangle = |i\rangle + z|i\rangle, \quad |\hat{i}\rangle = |i\rangle$
 $|\hat{j}\rangle = |j\rangle, \quad |j\rangle = |j\rangle - z|i\rangle$
 and for adjacent Gluons

$[-, -]$: -shift on $A_5[1^- 2^- 3^+ 4^+ 5^+]$
 $\rightarrow [1, 2]$ shift on $A_5[1^+ 2^+ 3^- 4^- 5^-]$

$$A_5 = \sum_I \sum_{u \neq I} \hat{A}_{2, I} \left(\frac{1}{z_I} \right) \tilde{A}_{2, I}(z_I)$$

$$= \sum_{k=4}^n \hat{A}_{2, k} \text{ (diagram with } \hat{P}_k \text{)} \text{ (diagram with } \hat{P}_k \text{)}$$

$$= \sum_{k=4}^n \sum_{u \neq k} \hat{A}_{2, k} \text{ (diagram with } \hat{P}_k \text{)} \text{ (diagram with } \hat{P}_k \text{)}$$

$$= \sum_{u \neq n} \sum_{k=4}^n \hat{A}_{n-k+2} [1^-, \hat{P}_k^h, u^+, \dots, n^+] \frac{2}{P_{\pm}^2}$$

$$P_I = P_2 + \dots + P_{n-1}$$

$$\hat{P}_I = \hat{P}_2 + \dots + \hat{P}_{n-1}$$

* Sum over n_I :

$$\sum_{k=0} \left(\hat{\Lambda}_{n-k+2} [1^-, \hat{P}_I^+, k^+, \dots, n^+] \frac{1}{P_I^2} \hat{\Lambda}_{n-1} [-\hat{P}_I^{-1}, \hat{2}^-, 3^-, \dots, (n-1)^+] \right.$$

$$\left. + \hat{\Lambda}_{n-k+2} [1^-, \hat{P}_I^+, k^+, n^+] \frac{1}{P_I^2} \hat{\Lambda}_{n-1} [-\hat{P}_I^+, 2^-, 3^-, \dots] \right)$$

* First term: $\hat{\Lambda}_{n-k+2}$ on at most be order 3:

$$n - k + 3 = 3$$

$$(n = k) \Rightarrow$$

(1st):

$$\hat{\Lambda}_3 [1^-, \hat{P}_I^+, n^+] \frac{1}{P_I^2} \hat{\Lambda}_{n-2} [-\hat{P}_I^{-1}, 2^-, 3^-, \dots, (n-1)^+]$$

where $n^+ = 5$, as the example

$$= \hat{\Lambda}_3 [1^-, \hat{P}_I^+, 5^+] \frac{1}{P_I^2} \hat{\Lambda}_4 [-\hat{P}_I^{-1}, 2^-, 3^-, 4^+]$$

variables:

$$(1s^4) = 0$$

Second term:

$$\sum_{k=0}^5 \hat{\Lambda}_{n-k} [1^-, \hat{P}_I^-, k^+, s^+] \frac{1}{P_I^2} \hat{\Lambda}_{n-1} [-\hat{P}_I^+, 2^-, 3^-]$$

$$= \hat{\Lambda}_4 [1^-, \hat{P}_I^-, 4^+, s^+] \frac{1}{P_I^2} \hat{\Lambda}_3 [-\hat{P}_I^+, 2^-, 3^-]$$

$$+ \hat{\Lambda}_5 [1^-, \hat{P}_I^-, s^+] \frac{1}{P_I^2} \hat{\Lambda}_4 [-\hat{P}_I^+, 2^-, 3^-, 4^+]$$

$$\hat{2}^- \quad \hat{P}_{23} \quad \hat{2}^- \quad \hat{P}_{45} = \hat{P}_i + \hat{P}_j$$



Claim: $A_3 [-\hat{P}_{23}^+, \hat{2}^-, 3^-] = 0$

Proof:-

$$A_3 = \frac{\langle \hat{2}^-, 3^- \rangle^3}{\langle 3 | \hat{P}_{23}^+ \rangle \langle \hat{P}_{23}^+ | \hat{2}^- \rangle}$$

$$[\alpha \hat{P}_{23}] \langle \hat{P}_{23}^+ | 3^- \rangle = \hat{P}_2 + P_3 | 3^- \rangle$$

$$= \hat{P}_2 | 3^- \rangle$$

$$[\alpha | \hat{2}^-] \langle 23 \rangle$$

$$= \left(\frac{\langle 23 \rangle^3 [\alpha \hat{P}_{23}]}{[\alpha \hat{2}^-] \langle \hat{2}^- \rangle} \right) \left(\frac{[\beta \hat{P}_{23}]}{\langle 23 \rangle [\beta 3^-]} \right)$$

$$\hat{P}_{23} = \hat{P}_2 \cdot P_3 = \langle \hat{2}^- | 3^- \rangle [23] = 0$$

$$\boxed{\langle \hat{2}^- | 3^- \rangle = 0}$$

So, first diagram vanishes.

Left with

$$A_5 [1^-, 2^-, 3^-, 4^+, 5^+] =$$

$$= A_3 [\hat{1}^-, \hat{P}_{1s}^+, 5^+] \frac{1}{P_{1s}^2} A_4 [-\hat{P}_{1s}^-, \hat{2}^-, 3^+, 4^+]$$

$$= \left(\frac{\langle 1 | \hat{P} \rangle^3}{\langle \hat{P} 5 \rangle \langle 5^+ | \hat{1}^- \rangle} \right) \left(\frac{1}{\langle 1s \rangle \langle 1s \rangle} \right) \left(\frac{\langle \hat{2}^- | 3^+ \rangle^4}{\langle \hat{2}^- | 3^+ \rangle \langle 3^+ | 4^+ \rangle \langle 4^+ | \hat{P} \rangle} \right)$$

• M.M. by $[\hat{p}_x]/[\hat{p}_x]^3$, $[\hat{p}_x] \neq 0$

Choose $x = 2$:

$$\bullet \langle \hat{1} \hat{p} \rangle [\hat{p}_2] = - \langle \hat{1} | \hat{1} + 5/2 \rangle = - \langle \hat{1} | 5 | 2 \rangle$$

$$= \langle \hat{1} 5 \rangle [5 2]$$

$$\bullet \langle 5 \hat{p} \rangle [\hat{p}_2] = \langle 5 1 \rangle [1 2]$$

$$\bullet \langle 4 \hat{p} \rangle [\hat{p}_2] = + \langle 4 | (\hat{2} + 3 + 4) | 2 \rangle$$

$$= - \langle 4 3 \rangle [3 2]$$

$$\bullet \langle \hat{2} \hat{p} \rangle [\hat{p}_2] = - 2 \hat{p}_2 \cdot \hat{p}$$

$$= 2 \tilde{p}_2 \cdot (\hat{p}_2 + p_3 + p_4)$$

$$= 2 [(\hat{p}_2 + p_3 + p_4) - (p_3 + p_4)] \cdot [\tilde{p}_2 + p_3 + p_4]$$

$$= 2 (\hat{p}_2 + p_3 + p_4)^2 - 2 (p_3 + p_4) \cdot \hat{p}_2$$

$$- 2 (p_3 + p_4)^2$$

$$\tilde{p}_2 \cdot (p_3 + p_4) = (p_2 + p_3 + p_4)^2 - (p_3 + p_4)^2$$

$$= - \langle 3 4 \rangle [3 4]$$

$$\langle \hat{2} \hat{p} \rangle [\hat{p}_2] = - \langle 3 4 \rangle [3 4]$$

$$\frac{1}{\langle \hat{2} 3 \rangle \langle 3 4 \rangle} \left(\frac{(\langle 1 5 \rangle [5 2])^3}{\langle 5 1 \rangle [1 2] \langle 5 \hat{1} \rangle} \right) \left(\frac{1}{\langle 1 5 \rangle [1 5]} \right) \left(\frac{\langle \hat{2} 3 \rangle^4}{\langle 3 4 \rangle [2 3]} \right)$$

$\uparrow$
 (-1)

$(= \langle 3 4 \rangle [3 4])$

$$= \frac{[5 2]^3 \langle \hat{2} 3 \rangle^3 \langle 1 5 \rangle^2}{\langle 3 4 \rangle [1 2] \langle 5 \hat{1} \rangle \langle 1 5 \rangle [1 5] \langle 3 4 \rangle [2 3] \langle 3 4 \rangle [3 4]}$$

$$= \frac{[2 5]^3 \langle \hat{2} 3 \rangle^3}{[1 2] [2 3] [3 4] [1 5] \langle 3 4 \rangle^3}$$

• Evaluate at residue $z = z_{15}$
 s.t. $\hat{p}_{15} = 0$

$$0 = \hat{P}_{15}^2 = \langle 15 | \hat{P}_{15}^2 | 15 \rangle$$

$$0 = \langle 15 | \hat{P}_{15}^2 | 15 \rangle = \langle 15 | \hat{P}_{15} | 15 \rangle + \langle 15 | \hat{P}_{15} | 25 \rangle$$

$$\hat{P}_{15} = - \frac{\langle 15 |}{\langle 25 |}$$

Thus,

$$\langle 15 | \hat{P}_{15} | 15 \rangle = \langle 15 | \hat{P}_{15} | 25 \rangle + \frac{\langle 15 |}{\langle 25 |} \langle 15 |$$

$$= \langle 15 | \hat{P}_{15} | 25 \rangle + \frac{\langle 15 |}{\langle 25 |} \langle 15 |$$

$$\langle 15 | \hat{P}_{15} | 15 \rangle = \langle 15 | \hat{P}_{15} | 25 \rangle + \frac{\langle 15 |}{\langle 25 |} \langle 15 |$$

$$= \frac{\langle 15 | \hat{P}_{15} | 25 \rangle}{\langle 25 |}$$

$$\langle 15 | \hat{P}_{15} | 15 \rangle = \frac{\langle 15 | \hat{P}_{15} | 25 \rangle}{\langle 25 |}$$

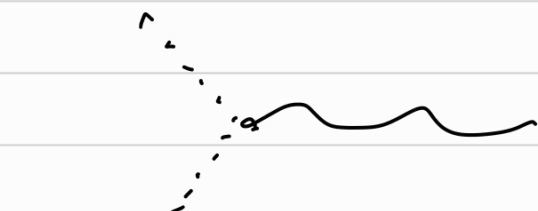
$$A_5 [1^- 2^- 3^- 4^+ 5^+] = \frac{[25]^3}{[12][23][34][45][51]} \left(\frac{[45][51]^3}{[25]^3} \right)$$

$$= \frac{[45]^4}{[12][23][34][45][51]}$$

ex 3.5 | Scalar QED

$$\Delta_3 (\phi \phi^* \gamma^\pm) = \langle 12 \rangle^4 \langle 23 \rangle^4 \langle 31 \rangle^4$$

$$= \langle 12 \rangle^{\pm 1} \langle 23 \rangle^{\mp 1} \langle 31 \rangle^{\mp 1}$$



Use $[4, 3]$ - shift on $A_\mu (\phi \phi^* \gamma^+ \gamma^-)$

$$\sum_{\tilde{3}^-} \hat{P}_L (R) \gamma^+$$

$$= \begin{array}{c} \text{diagram with } u^- \text{ and } \hat{p}_{31} \end{array} + \begin{array}{c} \text{diagram with } \hat{p}_{32} \end{array}$$

$$\hat{p}_{31} = \hat{p}_3 + p_1$$

$$A_3[1^0, \hat{p}_{31}^+, 4^-] \left| \frac{1}{p_{u1}} \right| A_3[2^0, \hat{p}_3^+, -\hat{p}_{u1}^-] \\ + \dots \langle 1 | \hat{p}_{u1} \rangle^{-1-1} \langle \hat{p}_3, 4 \rangle^{0-1+1} \langle 4 | \rangle^{-1}$$

NOTE: The Sum $\sum_{u_I}$ is over all possible spins of the theory. As such, for scalar QED $u_I \in \{+1, -1, 0\}$
 $\Rightarrow u_I = \pm 1$ must vanish: there is no possible way to make $[A_3] \sim m^1$
 It is always either m^2 or m^{-2}

$$A_4(\phi\phi^*\gamma^+\gamma^-) = \hat{A}_3[3^+, \hat{p}_{23}^0, 2^0] \frac{1}{p_{23}^2} A_3[-p_{23}^0, 4^+, 1^0] \\ = g^2 \frac{\langle 1u \rangle \langle 2u \rangle}{\langle 13 \rangle \langle 23 \rangle}$$

ex 3.6 Calculate $M_4(1^- 2^- 3^+ 4^+)$

$\rightarrow$ Employ $[1, 2]$ -shift $p_I = \hat{2} + 3$

$$M_4(1^- 2^- 3^+ 4^+) = \sum_{u_I} M_3(\hat{1}^-, \hat{p}_I^{u_I}, 4^+) \frac{1}{p_I} M_3(-\hat{p}_I^{u_I}, \hat{2}^-, 3^+)$$

$$M_3(1^-, \hat{p}^+, 4^+) = \langle 1 | \hat{p} \rangle^{2-2+2} \langle \hat{p} | 4 \rangle^{-2+2-2} \langle 4 | \rangle^{2+2-2} \\ = k^{-1} \frac{\langle 1 | \hat{p} \rangle^2 \langle 4 | \rangle^2}{\langle \hat{p} | 4 \rangle^6}$$

$$[k^{-1}] = m^{-1}$$

thus, this is the wrong signs, we should have

$$M_3(\hat{1}^-, \hat{p}_I^+, 4^+) = K^{-1} \frac{[\hat{p}_4]^6}{[\hat{1}\hat{p}]^2 [41]^2}$$

$$\hat{p}_{23} = \langle \hat{2}3 \rangle [\hat{2}3] = 0$$

$$\langle \hat{2}3 \rangle = 0$$

$$M_3(-\hat{p}_I^-, \hat{2}^-, 3^+) = \frac{\langle \hat{2}3 \rangle^6}{} = 0$$

$$\therefore \underline{h_I = -1}$$

$$M_3(\hat{1}^-, \hat{p}_I^-, 4^+) = \left(\frac{1}{K}\right) \frac{\langle \hat{1}\hat{p} \rangle^6}{[14]^2 \langle \hat{p}4 \rangle^2}$$

$$M_3(-\hat{p}_I^+, \hat{2}^-, 3^+) = \left(\frac{1}{K}\right) \frac{[3\hat{p}]^6}{[3\hat{2}]^2 [\hat{p}2]^2}$$

$$K^{-2} \left(\frac{\langle \hat{1}\hat{p} \rangle^6}{\langle 14 \rangle^2 \langle \hat{p}4 \rangle^2} \right) \left(\frac{1}{\langle 23 \rangle [23]} \right) \left(\frac{[3\hat{p}]^6}{[3\hat{2}]^2 [\hat{p}\hat{2}]^2} \right)$$

$$\left(\frac{\langle \hat{1}\hat{p} \rangle [\hat{p} 3]}{\langle 12 \rangle^6 [\hat{2} 3]^4} \right)^6 = \left(\langle 1\hat{2} \rangle [\hat{2} 3] \right)^6$$

$$= \frac{\langle 12 \rangle^6 [\hat{2} 3]^6}{\langle 14 \rangle^2 \langle 23 \rangle [23] \left(\langle 4\hat{p} \rangle [\hat{p}\hat{2}] \right)^2}$$

$$= \frac{\langle 12 \rangle^6 [23]^3}{2[4]^2 \langle 23 \rangle (\langle 43 \rangle [32])^2} K^{-2}$$

$$= \frac{\langle 12 \rangle^7 (-[32])}{\langle 14 \rangle^2 \langle 23 \rangle \langle 34 \rangle^2 \langle 12 \rangle}$$

$$\left[\begin{array}{l} \langle 4 | \hat{1} \rangle [\hat{1} | \hat{2}] = - \langle 4 | 3 \rangle [3 \hat{2}] \\ \therefore \frac{-[3\hat{2}]}{\langle \hat{1}4 \rangle} = \frac{[\hat{1}\hat{2}]}{\langle 43 \rangle} \end{array} \right]$$

$$= \frac{\langle 12 \rangle \langle 12 \rangle}{\langle 14 \rangle \langle 43 \rangle \langle 23 \rangle \langle 12 \rangle \langle 34 \rangle^2}$$

$$|1\rangle \langle 23| + |2\rangle \langle 31| + |3\rangle \langle 12| = 0$$

$$\langle 41| \langle 23| + \langle 42| \langle 31| + \langle 43| \langle 12| = 0$$

$$\langle 12| = \langle 12|$$

$$\frac{\langle 12 \rangle^7 \langle 12 \rangle}{\langle 14 \rangle (-\langle 24 \rangle \langle 13 \rangle) \langle 23 \rangle \langle 34 \rangle^2}$$

Close... Up to a - sign and

$\langle 24 \rangle^{-1}$ instead of $\langle 24 \rangle^{-1}$

$$M_4(1^- 2^- 3^+ 4^+) = \frac{\langle 12 \rangle^7 \langle 12 \rangle}{\langle 13 \rangle \langle 14 \rangle \langle 23 \rangle \langle 24 \rangle \langle 34 \rangle^2}$$

* Note M_4 has Bose symmetry under $1 \leftrightarrow 2$ and $3 \leftrightarrow 4$

Claim: $M_4(1^- 2^- 3^+ 4^+)$ Obeys the 4pt KLT-relations

$$M_4(1234) = -s_{12} A_4[1234] A_4[1243]$$

Proof: $A_4[1^- 2^- 3^+ 4^+] = \frac{\langle 12 \rangle^4}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$

$$\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle$$

$$\therefore A_4[1234] A_4[1243]$$

$$= \left(\frac{\langle 12 \rangle^8}{\langle 12 \rangle \langle 24 \rangle \langle 43 \rangle \langle 31 \rangle \langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

and so

$$-s_{12} = (p_1 + p_2)^2 = \langle 12 \rangle [12]$$

$$= \frac{\langle 12 \rangle^7 [12]}{\langle 24 \rangle \langle 43 \rangle \langle 31 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$$



* Split-helicity NMHV: $A_6[1^- 2^- 3^- 4^+ 5^+ 6^+]$

Take a $[1, 2^-]$ -shift /

$$A_6[] = \sum_{k=4}^6 \sum_{h_I} \hat{A}_L(z_I) \frac{1}{P_I^2} \hat{A}_R(z_I)$$

$$= \sum_{h_I = \pm 1} \sum_{k=4}^6 \hat{1} \text{---} (L) \text{---} \hat{P}_I^{h_I} (R) \text{---} \hat{2} \text{---} 3^+ \text{---} (k-1)^+$$

$$= \sum_{h_I} \sum_{k=4}^6 \hat{A}_{q-k}[\hat{1}^-, \hat{P}_I^{h_I}, k^+, \dots, 6] \frac{1}{P_I^2}$$

$$\hat{A}_{k-1}[-\hat{P}_I^{-h_I}, \hat{2}^-, 3^+, \dots, (k-1)^+]$$

$$= \sum_{h_I} \left\{ \hat{A}_5[\hat{1}^-, \hat{P}_I^{h_I}, 4^+, 5^+, 6^+] \frac{1}{P_I^2} \right.$$

$$\textcircled{a} \quad \left. \hat{A}_3[-\hat{P}_I^{h_I}, \hat{2}^-, 3^+] \right.$$

$$\textcircled{b} \quad + \hat{A}_4[\hat{1}^-, \hat{P}_I^{h_I}, 5^+, 6^+] \frac{1}{P_I^2}$$

$$\hat{A}_4[-\hat{P}_I^{h_I}, \hat{2}^-, 3^+, 4^+]$$

$$\textcircled{c} \quad + \hat{A}_3[\hat{1}^-, \hat{P}_I^{h_I}, 6^+] \frac{1}{P_I^2} \hat{A}_5[-\hat{P}_I^{h_I}, \hat{2}^-, 3^+, 4^+, 5^+]$$

$\textcircled{a}$ Vanishes if $h_I = +1$ and not if $h_I = -1$

$\textcircled{b}$ Vanishes if $h_I = +1$ and not if $h_I = -1$

$\textcircled{c}$ Both $\pm 1 = h_I$ seem fine

$\textcircled{a}$ I believe $[- - -]$ is unphysical i.e.

the engineering dim is incorrect.

$$A_3[1^- 2^- 3^-] = \langle 12 \rangle \langle 23 \rangle \langle 31 \rangle$$

$$\text{So } [A_3] \sim m^4$$

BA RHS $\sim m^3$ i.e. it is unphysical

$$\therefore \boxed{\textcircled{A}} = 0$$

For both $h_{\pm} = \pm 1$

(b) We obtain the term $P_L = P_2 \dots P_{k-1}$

$$\hat{A}_u[1^-, \hat{P}_{156}^-, 5^+, 6^+] \frac{1}{P_{156}^2} A_u[-\hat{P}_{156}^+, \hat{2}^-, 3^-, 4^+]$$

(c) $A_3[1^-, 2^+, 3^+] \sim [M]^{-1}$

which is the unphysical mass dim.

Thus, $h_{\pm} = -1$

$$\hat{A}_3[1^-, \hat{P}_{16}^-, 6^+] \frac{1}{P_{16}^2} A_5[-\hat{P}_{16}^+, 5^-, 3^-, 4^+, 5^+]$$

Thus, we are left with

$$A_6[1^- 2^- 3^- 4^+ 5^+ 6^+] =$$

(B) has a 3-particle pole $\hat{P}_{156}$ from $(P_{156}^2)^{-1}$

• would expect no distinction between

the $(-++)$ 3-particle channels 561 and 345. Thus, we would expect to see a pole at $P_{345}^2 = P_{126}^2 = 0$ (i.e. helicity flip).
 * The $P_{345}^2 = P_{126}^2 = 0$ pole is actually HIDDEN in the denominator factor $\langle \hat{2} \hat{P}_{16} \rangle$

of diagram (A).

$$\hat{A}_3[\hat{1}^-, \hat{P}_{16}^-, 6^+] \frac{1}{P_{16}^2} A_5[-\hat{P}_{16}^+, \hat{2}^-, 3^+, 4^+, 5^+]$$

$$A_5[\hat{2}^-, 3^+, 4^+, 5^+, -\hat{P}_{16}^+] = \frac{\langle \hat{2} 3 \rangle^4}{\langle \hat{2} 3 \rangle \langle 3 4 \rangle \langle 4 5 \rangle \langle 5 \hat{P} \rangle \langle \hat{P} \hat{2} \rangle}$$

$$\textcircled{A} = \left(\frac{\langle \hat{1} \hat{P} \rangle^3}{\langle \hat{P} 6 \rangle \langle 6 1 \rangle} \right) \left(\frac{1}{\langle 16 \rangle \langle 16 \rangle} \right) \left(\frac{\langle \hat{2} 3 \rangle^4}{\langle \hat{2} 3 \rangle \langle 3 4 \rangle \langle 4 5 \rangle \langle 5 \hat{P} \rangle \langle \hat{P} \hat{2} \rangle} \right)$$

Mult. by $[\hat{P}_{16} 3] / [\hat{P}_{16} 3]$

$$\Rightarrow \langle \hat{2} \hat{P} \rangle [\hat{P} 3] = \langle 2 1 \rangle [1 3] + \langle \hat{2} 6 \rangle [6 3]$$

Since

$$\hat{P}_{16}^2 = 0 \Rightarrow [\hat{1} 6] = 0$$

$$[1 6] + z[2 6] = 0 \Rightarrow z_{16} = -\frac{[1 6]}{[2 6]}$$

as such,

$$\begin{aligned} \langle \hat{2} 6 \rangle &= \langle 2 6 \rangle + \frac{[1 6]}{[2 6]} \langle 1 6 \rangle \\ &= \frac{1}{[2 6]} ([2 6] \langle 2 6 \rangle + [1 6] \langle 1 6 \rangle) \end{aligned}$$

$$[\hat{1} 3] = [1 3] - \frac{[1 6]}{[2 6]} [2 3] = \frac{1}{[2 6]} ([2 6] [1 3] - [1 6] [2 3])$$

$$[6 1] ([1 3] [2 3] + [2 3] [1 3] + [1 3] [1 2]) = 0$$

$$- [1 6] [2 3] + [6 2] [3 1] + [6 3] [1 2] = 0$$

$$\therefore () = - [6 3] [1 2] = [3 6] [1 2]$$

$$[\hat{1} 3] = \frac{[3 6] [1 2]}{[2 6]}$$

$$\frac{[3 6] [1 2]}{[2 6]} = 1$$

$$\begin{aligned}
 \bullet \langle \hat{2} \hat{p} \rangle [\hat{p} 3] &= \langle 21 \rangle ([26]) + [26] ([26] \langle 26 \rangle + [16] \langle 16 \rangle) [3] \\
 &= - \left(\frac{[36]}{[26]} \right) (\langle 12 \rangle [12] + \langle 26 \rangle [26] + [16] \langle 16 \rangle) \\
 &= - ([36]/[26]) P_{126}^2
 \end{aligned}$$

$$\textcircled{A} \sim \left(\begin{array}{c} -[26] \\ [36] P_{126}^2 \end{array} \right)$$

Thus, P_{126}^2 is also a pole!

$$\begin{aligned}
 \textcircled{A} &= \frac{[\hat{p} 3] \langle \hat{2} 3 \rangle^4 [26]}{[16] \langle \hat{2} 3 \rangle \langle 34 \rangle \langle 45 \rangle \langle 5 \hat{p} \rangle [36] P_{126}^2} \\
 &= \frac{[\hat{p} 3] \langle \hat{2} 3 \rangle^3 [26]}{[16] \langle 34 \rangle \langle 45 \rangle \langle 5 \hat{p}_{16} \rangle [36] P_{126}^2}
 \end{aligned}$$

$$\begin{aligned}
 \left\{ \begin{aligned}
 [\hat{p} 3] &= [13] + [63] = [13] - \frac{[16]}{[26]} [23] + [63] \\
 &= \frac{1}{[26]} ([13][26] - [16][23]) + [63] \\
 [13][26] + [123][63] + [16][32] &= 0 \\
 &= \frac{1}{[26]} ([12][36]) - [36] = \frac{[36]}{[26]} ([12] - [26]) \\
 &= \frac{[36]}{[26]} [\hat{p} 2]
 \end{aligned} \right.$$

$$\langle \hat{2} 3 \rangle = \langle 23 \rangle + \frac{[16]}{[26]} \langle 13 \rangle = \frac{1}{[26]} (\langle 31(1+2)16 \rangle)$$

$$\therefore \langle \hat{2} 3 \rangle^3 = \frac{(\langle 31(1+2)16 \rangle)^3}{[26]^3}$$

$$= \left(\frac{[\hat{p} 3] (\langle 31(1+2)16 \rangle)^3}{[16] \langle 34 \rangle \langle 45 \rangle \langle 5 \hat{p}_{16} \rangle [26]^2 [36] P_{126}^2} \right) \left(\frac{[\hat{p}_{16} 2]}{[\hat{p}_{16} 3]} \right)$$

$$\langle 5 \hat{p}_{16} \rangle [\hat{p}_{16} 2] = - \langle 5 | (1+6) | 2 \rangle$$

$$\leadsto \text{NO PE} = \text{Mult. By } \frac{[\hat{p}_{16} 3]^3}{[\hat{p}_{16} 3]^3}$$

$$\text{Denom: } \langle \hat{2} \hat{p} \rangle [\hat{p} 3] = - \frac{[36]}{[26]} P_{126}^2$$

$$\begin{aligned}
 -\langle 6\hat{p} \rangle [\hat{p} 3] &= -\langle 6\uparrow \rangle [\uparrow 3] + \cancel{0} \\
 &= -\langle 61 \rangle [\uparrow 3] \\
 &= \langle 16 \rangle [\uparrow 3] \quad \leftarrow
 \end{aligned}$$

$$\begin{aligned}
 \langle 5\hat{p} \rangle [\hat{p} 3] &= \langle 51 \rangle [\uparrow 3] + \langle 56 \rangle [6 3] \\
 &= \langle 51 \rangle [\uparrow 3] - \frac{[16]}{[26]} [2 3] + \langle 56 \rangle [6 3] \\
 &= \frac{[51]}{[26]} ([2 6] [\uparrow 3] - [1 6] [2 3]) + \langle 56 \rangle [6 3]
 \end{aligned}$$

$$\begin{aligned}
 &\left\{ \begin{aligned} 11[23] + 12[31] + 13[12] &= 0 \\ 61[23] + 62[31] + 63[12] &= 0 \\ -16[23] + 26[13] &= 36[12] \end{aligned} \right. \\
 &= \left(\frac{[51]}{[26]} 36[12] \right) + \langle 56 \rangle [6 3] \left(\frac{[26]}{[26]} \right) \\
 &= [26]^{-2} (\langle 51 \rangle 36[12] + \langle 56 \rangle [6 3] [26]) \\
 &= \frac{[36]}{[26]} (\langle 51 \rangle [12] - \langle 56 \rangle [26]) \\
 &= \frac{[36]}{[26]} (\langle 5 | (1+6) | 2 \rangle) \quad \neq
 \end{aligned}$$

$$\langle \hat{2} | = \langle 1 |$$

• Numerator

$$\begin{aligned}
 \langle \uparrow \hat{p} \rangle [\hat{p} 3] &= -\langle \uparrow | (\uparrow + 6 | 3 \rangle \\
 &= \langle \uparrow 6 \rangle [6 3]
 \end{aligned}$$

$$\therefore \left(\frac{(\langle \uparrow 6 \rangle [6 3])^3 \langle \hat{2} 3 \rangle^3}{\langle 3 4 \rangle \langle 4 5 \rangle \langle 1 6 \rangle [6 1] \langle \uparrow 6 \rangle} \right) \left(\frac{[26]}{[36] \langle 5 | (1+6) | 2 \rangle} \right)$$

$$\times \left(\frac{1}{\langle 1 6 \rangle [\uparrow 3]} \right) \times \left(\frac{-[26]}{[36] P_{126}^2} \right)$$

$$\begin{aligned}
 &\frac{[36] [26]^2 \langle \hat{2} 3 \rangle^3}{=} \\
 &= \langle 3 4 \rangle \langle 4 5 \rangle [6 1] \langle 5 | (1+6) | 2 \rangle [\uparrow 3] P_{126}^2
 \end{aligned}$$

$$= \left(\frac{[36]}{[26] [\uparrow 3]} \right) \left(\frac{(\langle 3 | (1+2) | 6 \rangle)^3}{\langle 4 5 \rangle \langle 3 4 \rangle [6 1] \langle 5 | (1+6) | 2 \rangle P^2} \right)$$

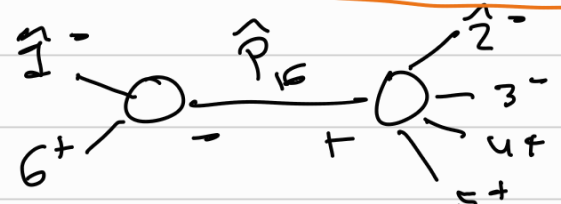
$$[\uparrow 3] = [13] - \frac{[16]}{[26]} [23] = \frac{1}{[26]} ([26][13] - [16][23])$$

$$= \frac{[36][12]}{[26]}$$

$$= \left(\frac{[36][26]}{[26][36][12]} \right) \left(\begin{array}{c} \\ \end{array} \right)$$

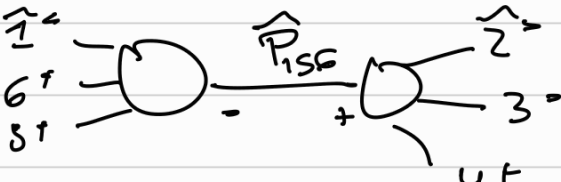
$$= \frac{(\langle 3|(1+2)|6 \rangle)^3}{P_{126}^2 [12][61] \langle 34 \rangle \langle 45 \rangle \langle 5|(1+6)|2 \rangle}$$

We thus conclude (A):

$$\therefore \hat{1}^- \xrightarrow{\hat{P}_{16}} \hat{2}^- \quad \langle 3|(1+2)|6 \rangle^3$$


$$= P_{126}^2 [21][16] \langle 34 \rangle \langle 45 \rangle \langle 5|(1+6)|2 \rangle$$

* Can Finally calculate (B)

$$\hat{1}^- \xrightarrow{\hat{P}_{156}} \hat{2}^-$$


$$\hat{A}_4(\hat{1}^-, \hat{P}_{156}, 6^+, 5^+) \frac{1}{P_{156}^2} A_4(-\hat{P}_{156}, \hat{2}^-, 3^-, 4^+)$$

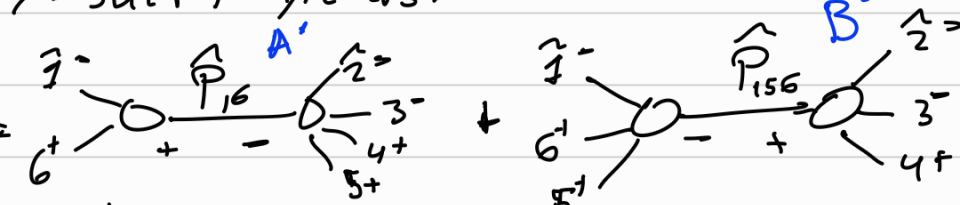
$$= \left(\frac{\langle \hat{1} \hat{P} \rangle^4}{\langle \hat{1} \hat{P} \rangle \langle \hat{P} 6 \rangle \langle 6 5 \rangle \langle 5 \hat{1} \rangle} \right) \left(\frac{1}{P_{156}^2} \right) \left(\frac{\langle \hat{2} 3 \rangle^4}{\langle \hat{2} 3 \rangle \langle 3 4 \rangle \langle 4 \hat{P} \rangle \langle \hat{P} 2 \rangle} \right)$$

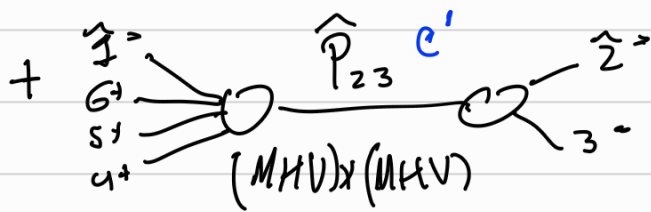
On-shell: $P_{156}^2 = 0 \Rightarrow z = - \frac{\alpha}{\langle 1|(5+6)|2 \rangle}$

$$\alpha \equiv \langle 15 \rangle [15] + \langle 16 \rangle [16] + [56] \langle 56 \rangle$$

oof... Quite long actually. Do at a later point....

* Using a $[2,1]$ -shift yields:

$$A_6[1^-, 2^-, 3^-, 4^-, 5^-, 6^+] = \text{(MHV)} \times \text{(MHV)} + \text{(MHV)} \times \text{(MHV)}$$




A' vanishes from 3-part. Kine metres

* BCFW relations are recursive both in particle number and in N^k MHV level K .

• $A + B = A' + B' + C'$ originate from residue theorem

~> Come Back and do ex: 3.9

* Physical poles occur when adjacent ext. lines go on-shell.

~> $[5|2+6|2]$ is a Spurious pole.

Locality is not manifest

$$\sim \oint_C \frac{\hat{A}_n(z_{\pm})}{z} = 0 \Leftrightarrow \text{Res}_{z=0}(\hat{A}_n) = -\sum_I \text{Res}_{z_I}(\hat{A}_n)$$

$$\text{Res}_{z=0}(\hat{A}_n) \equiv A_n = \sum_{\text{diagrams}} \hat{A}_1 \cdots \hat{A}_L \xrightarrow{\hat{P}_I} \hat{A}_R \xrightarrow{\hat{P}_I} \hat{A}_n$$

$$\equiv \sum_I d_I \quad \checkmark$$

Also have

$$\oint_C \hat{A}_n(z) = 0$$

$\Leftrightarrow$

$$\sum_I d_I z_I = 0 \leftarrow \text{Res}_{z=0}(\hat{A}) = 0$$

When Does it Work?:

(i) Yang-Mills theory and Gluon Scattering

- $A^2 \partial A$ and A^4

$\leadsto A^2 \partial A$ captures all into needed for all on-shell gluon amplitudes at tree-level i.e.

$$g A^2 \partial A$$

and $g^2 A^4$

So all higher order terms of $g A^2 \partial A$ yield into at g^2 .

(ii) Scalar-QED: $D_\mu = \partial_\mu - ieA_\mu$

$$\mathcal{L} \supset -|D\phi|^2 = |D\phi|^2 + ieA^\mu [\partial_\mu \phi^* \phi - \phi^* \partial_\mu \phi] - e^2 A^\mu A_\mu \phi^* \phi$$

- Role of e^2 is to ensure ^{an} off-shell gauge invariant Lagrangian.

However:

$$A_4^{BCFW}(\phi\phi^*\phi\phi^*) = \tilde{e}^2 \frac{\langle 13 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle}$$

But using Feynman Rules yields

$$A_4^{Feyn} = \tilde{e}^2 \left(1 + \frac{\langle 13 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

$\leadsto$ BCFW knows: 4d local theory w massless spin-1 particles and charged massless spin-0 and a dimensionless coupling.

For Full QED: $-\frac{1}{4}F_{\mu\nu}F^{\mu\nu} - |D\phi|^2 - \frac{1}{4}\lambda |\phi|^4$

$$A_4(\phi\phi^*\phi\phi^*) = -\lambda + \tilde{e}^2 \left(1 + \frac{\langle 13 \rangle^2 \langle 24 \rangle^2}{\langle 12 \rangle \langle 23 \rangle \langle 34 \rangle \langle 41 \rangle} \right)$$

Family of theories labelled by λ
no BCPW produces the special case of
 $\lambda = \tilde{e}^2$

* There is no way the 3pt interactions can know the
contents of λ/ϕ^4

(i) Scalar Theory: $\lambda\phi^4$

(ii) $N=4$ super Yang-Mills: 16 massless gluons $g^\pm$

4 Gluinos: λ^a, λ_a , helicity $h = \pm 1/2$ $a, b = 1, \dots, 4$

6 Scalars: S^{ab}

a, b : $SU(4)$ symmetry

(iii) Gravity: Tree level comp. determined by the
3-vertex graviton interaction.